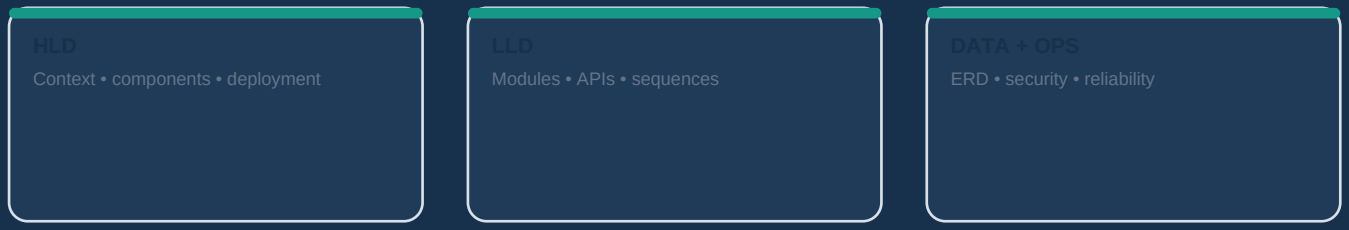
PRODUCT + ENGINEERING DESIGN

ROOM & ROOMMATE DISCOVERY MARKETPLACE

HLD • LLD • SYSTEM ARCHITECTURE

A clean visual guide to what the product does, how components connect, how data moves, and how developers should build it.



India-first • Single-city pilot • Modular monolith • No code listings

01 — How to read this blueprint

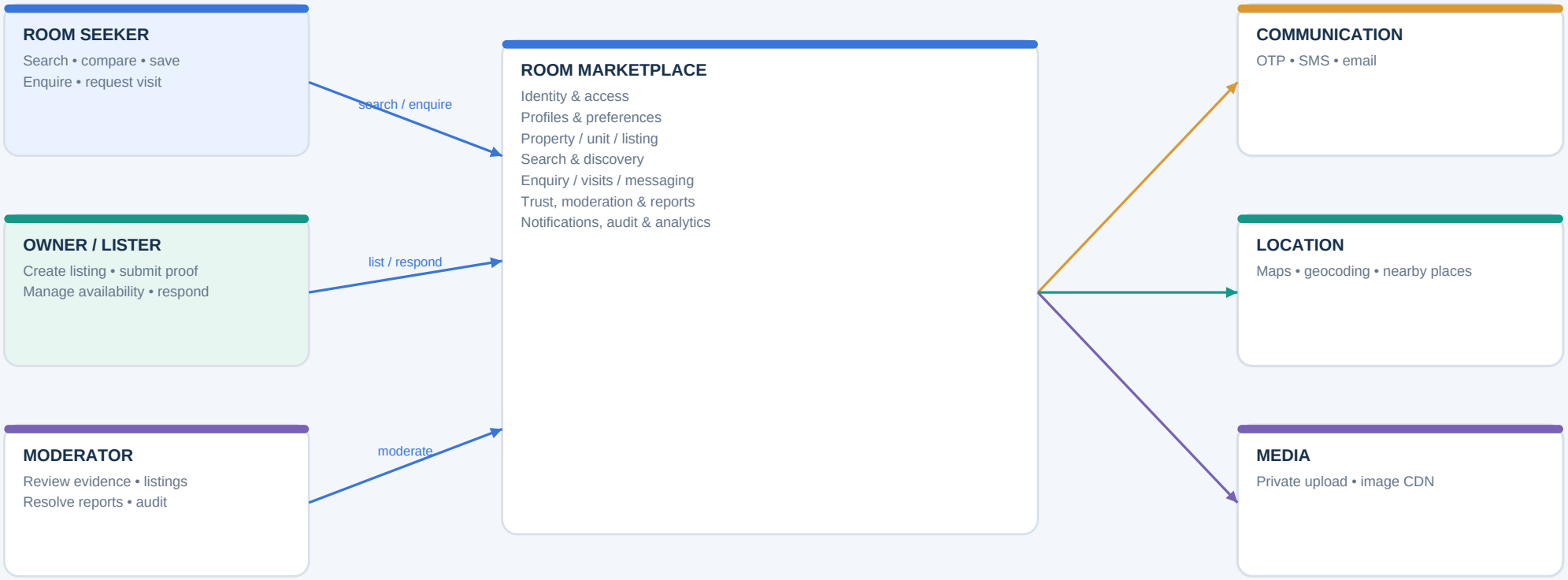
Follow the flow from product boundary → HLD → LLD → data design → production readiness.

1. **PRODUCT CONTEXT**
   Actors, use cases, scope and boundaries.
2. **HLD**
   Major components, system boundaries and deployment.
3. **LLD**
   Module responsibilities, API groups, sequences and state transitions.
4. **DATA DESIGN**
   Core entities, relationships, ownership and integrity.
5. **PRODUCTION**
   Security, reliability, testing, release and operations.

HLD answers: “What are the major parts and how do they connect?” LLD answers: “What happens inside each part, step by step?”

02 — Product context & system boundary

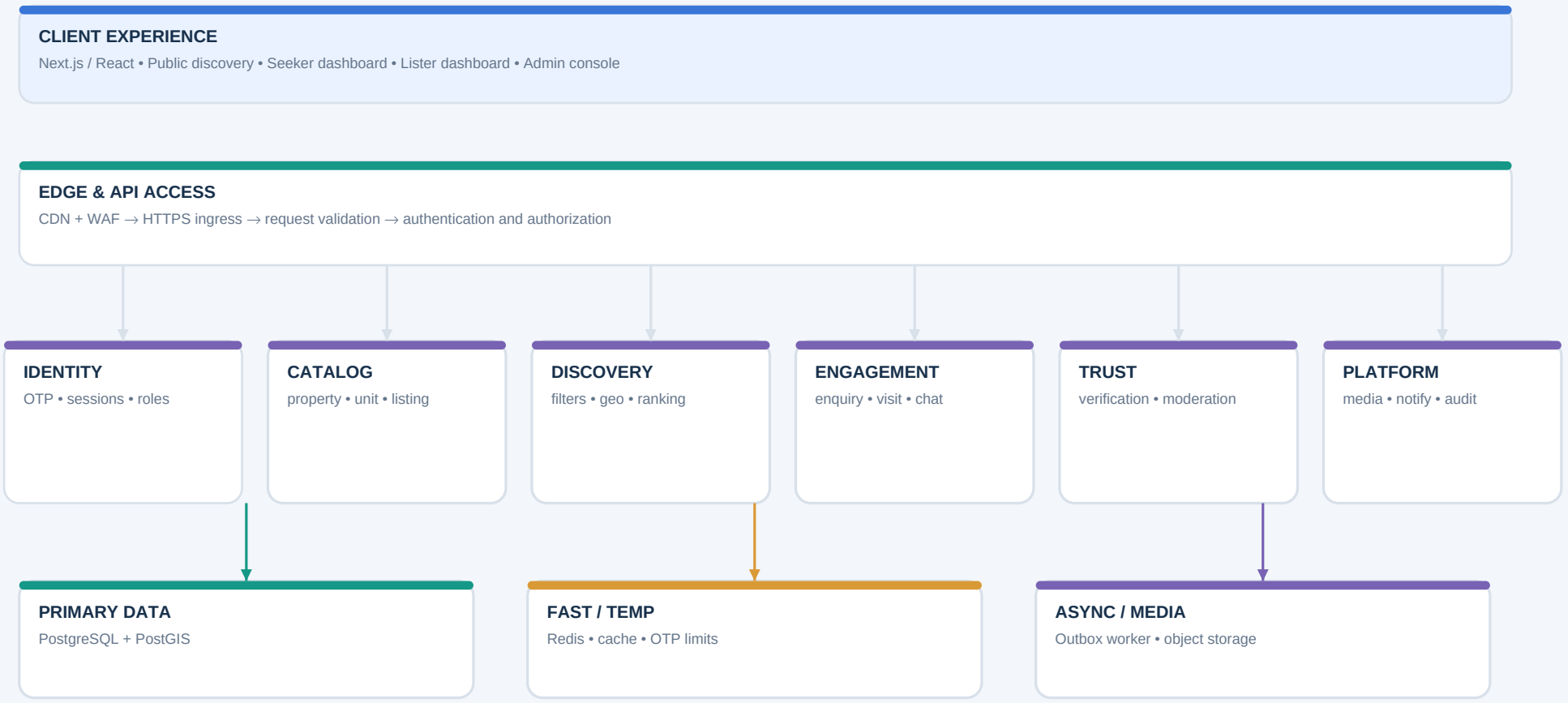
A discovery and lead-generation marketplace first; it is not the landlord, tenant or guarantor.



MVP excludes rent collection, escrow, automated criminality judgments, public exact-address exposure and opaque AI decisions.

03 — HLD: logical system architecture

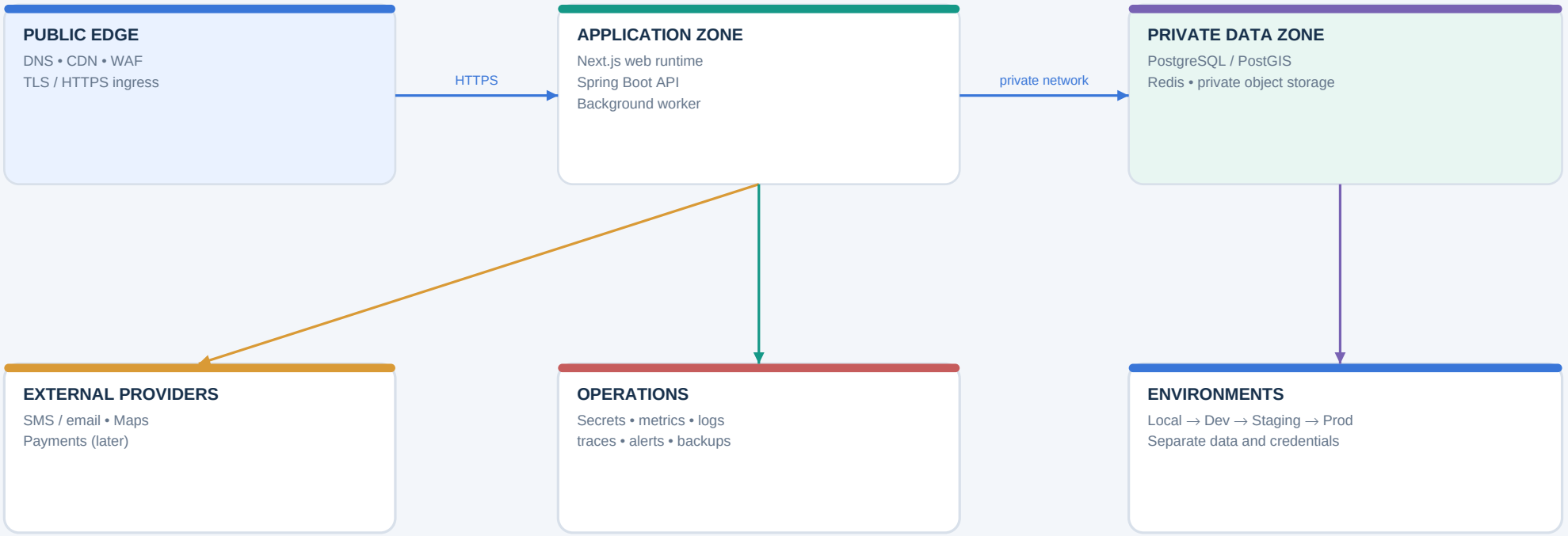
One web client and one modular backend deployment; infrastructure is separated by responsibility.



Architecture style: modular monolith. Modules are logically isolated in one Spring Boot application; extract services only when evidence justifies it.

04 — HLD: deployment architecture

Public traffic enters through protected edge services; data stores remain private.



No public database endpoint. Use least privilege, secret manager, private buckets, encryption at rest, tested backups and restore drills.

05 — HLD: backend module decomposition

Each module owns its use cases and data access; cross-module work uses interfaces or events.

IDENTITY & ACCESS

OTP, login/session, roles, account status
Owns identity and authorization

USER & PROFILE

Profile, preferences, consent, contact visibility
Owns profile data

PROPERTY CATALOG

Property, unit, inventory, listing, media metadata
Owns listing lifecycle

SEARCH & DISCOVERY

Text, filters, geo, sorting, explainable ranking
Reads published listing data

ENQUIRY & VISITS

Lead creation, visit slots, reschedule, cancellation
Owns interaction lifecycle

TRUST & MODERATION

Verification cases, reports, review decisions
Owns risk queues and decisions

MESSAGING & NOTIFY

Conversations, messages, notification delivery
Owns communication state

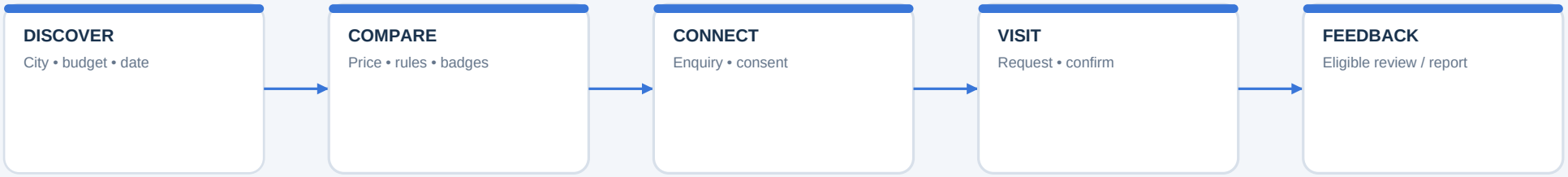
PLATFORM CORE

Audit, outbox, idempotency, configuration
Shared technical capabilities

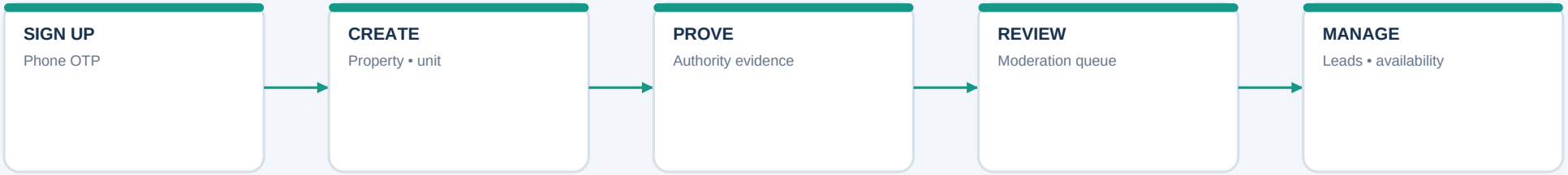
Boundary rule: one module must not directly write another module’s tables. Collaborate through an application service or asynchronous event.

06 — HLD: end-to-end user journeys

One account may act as seeker and lister; capabilities depend on permissions and resource ownership.



SEEKER FLOW

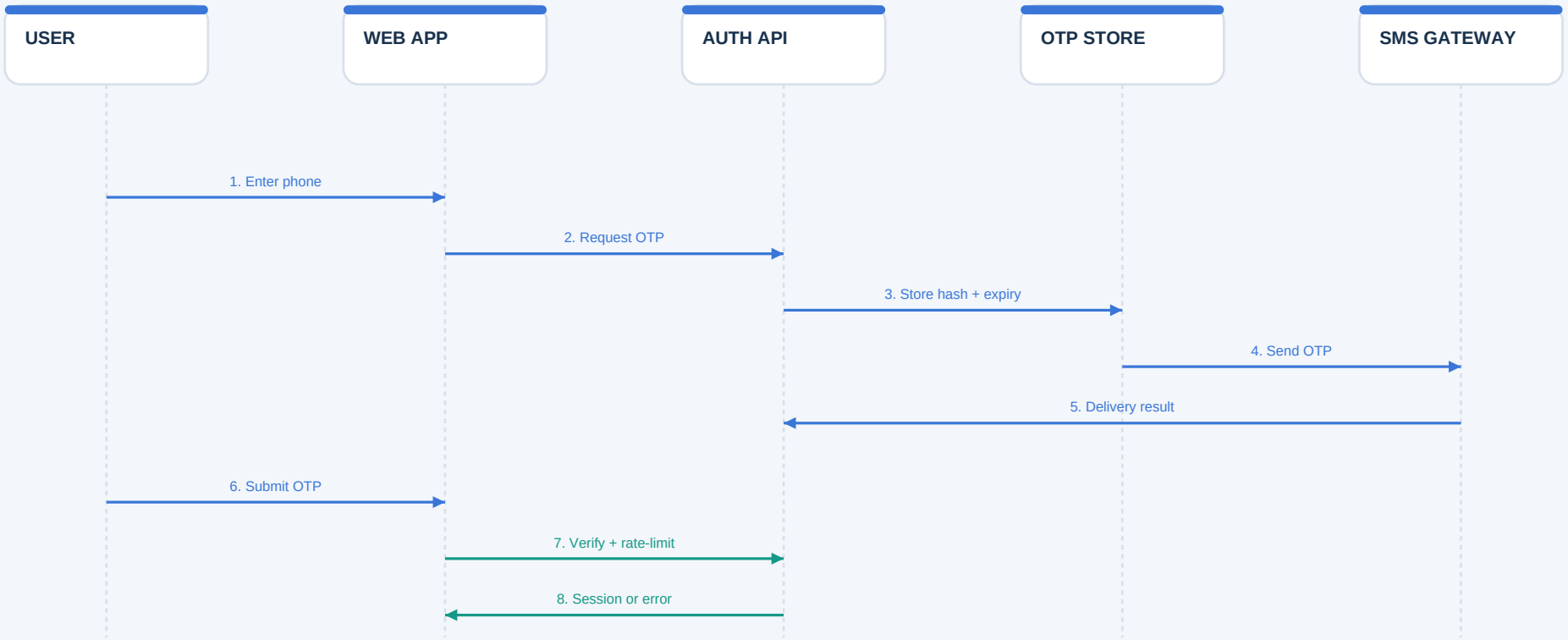


LISTER FLOW

A phone-verified account is not an owner-verified listing. Authority evidence and listing moderation are separate decisions.

07 — LLD: sign-in with phone OTP

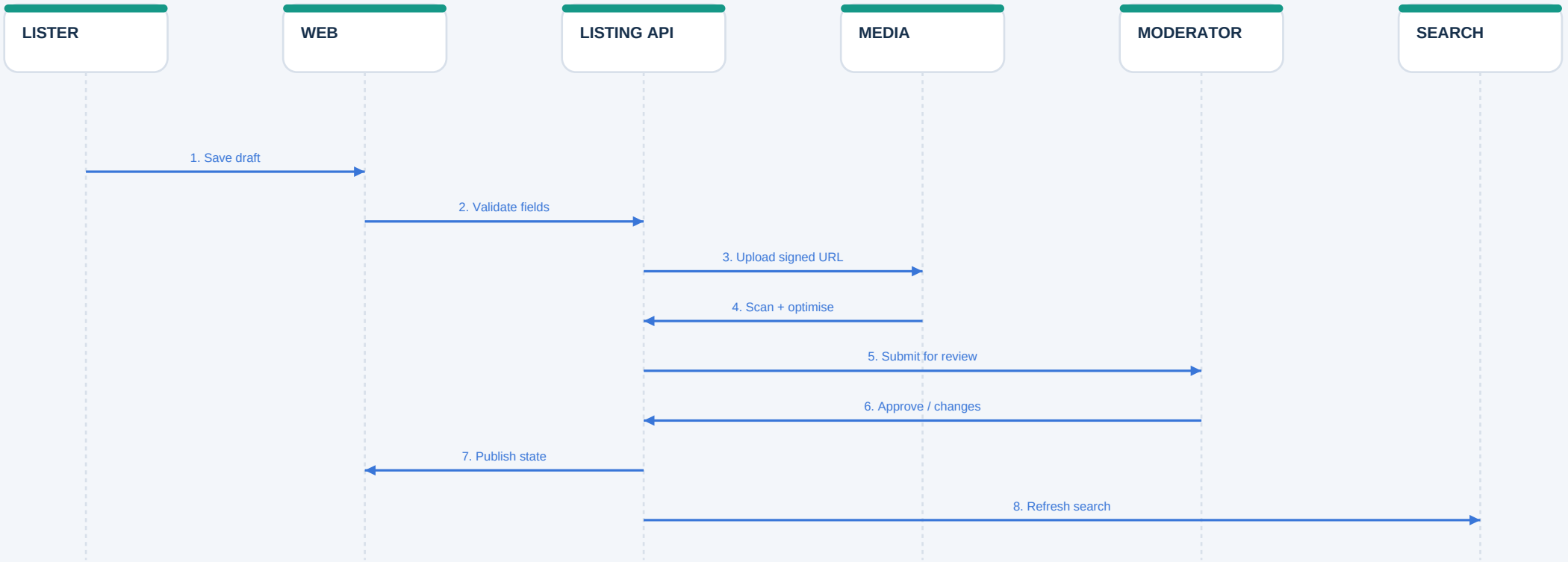
Provider sends the code; backend owns validation, attempt limits and session issuance.



OTP is short-lived and never logged. Store a hash, throttle by phone/IP/device, cap retries, invalidate after success and issue short-lived session credentials.

08 — LLD: create, review and publish a listing

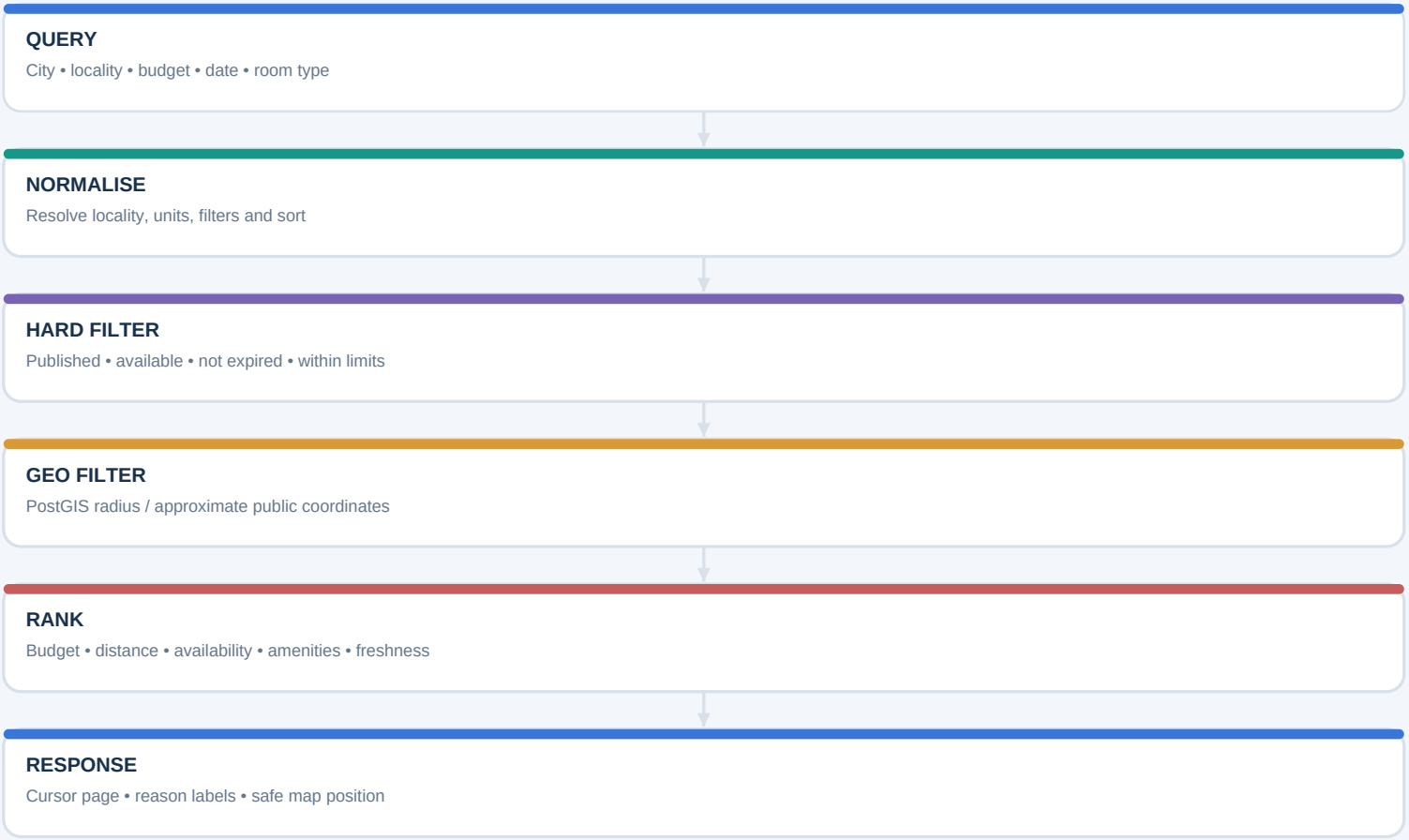
A listing is not searchable until validation and moderation both succeed.



Needs-changes / rejected listings remain private. Store moderator actor, timestamp, reason code and evidence reference for every decision.

09 — LLD: search and discovery pipeline

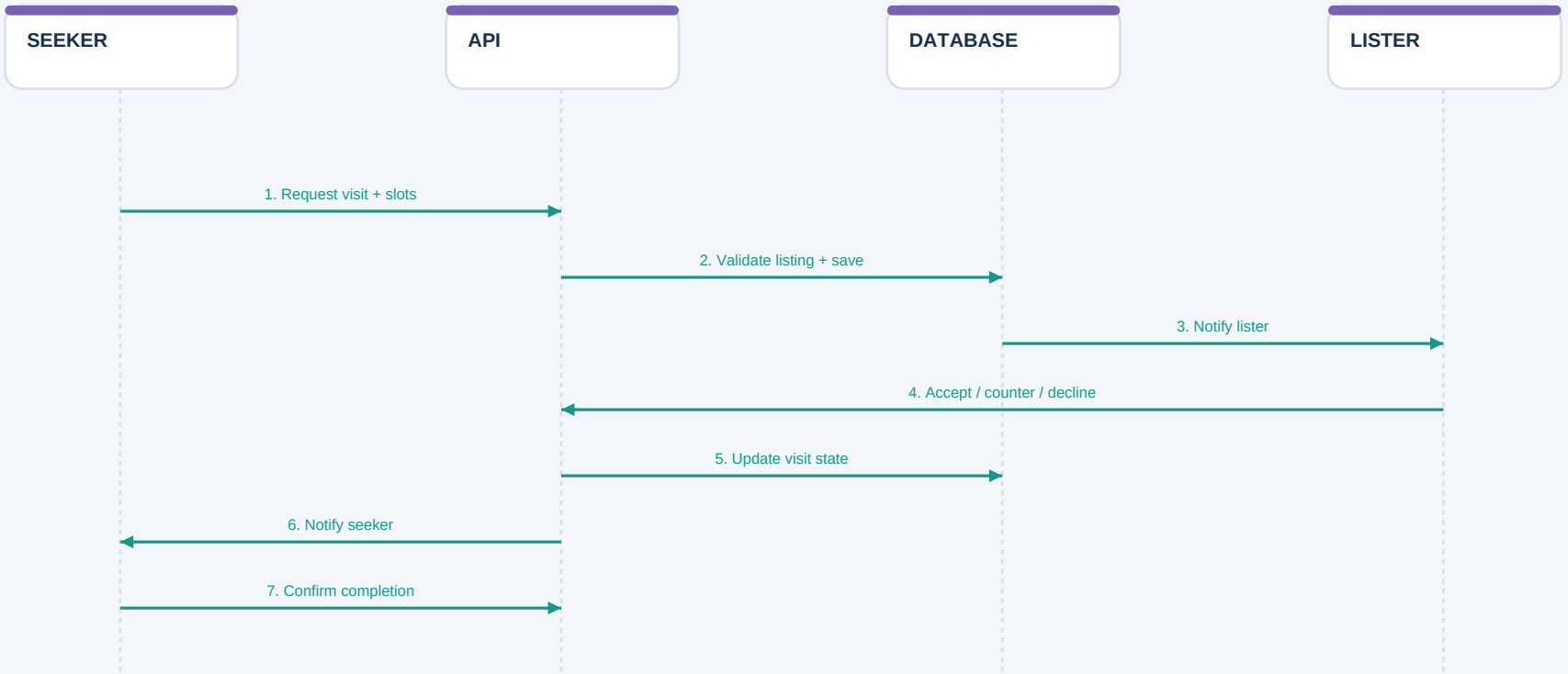
Hard constraints first; relevance ranking comes only after unavailable and ineligible listings are removed.



Start with deterministic, explainable ranking. AI recommendations are a later opt-in layer; do not infer sensitive traits.

10 — LLD: enquiry and visit scheduling

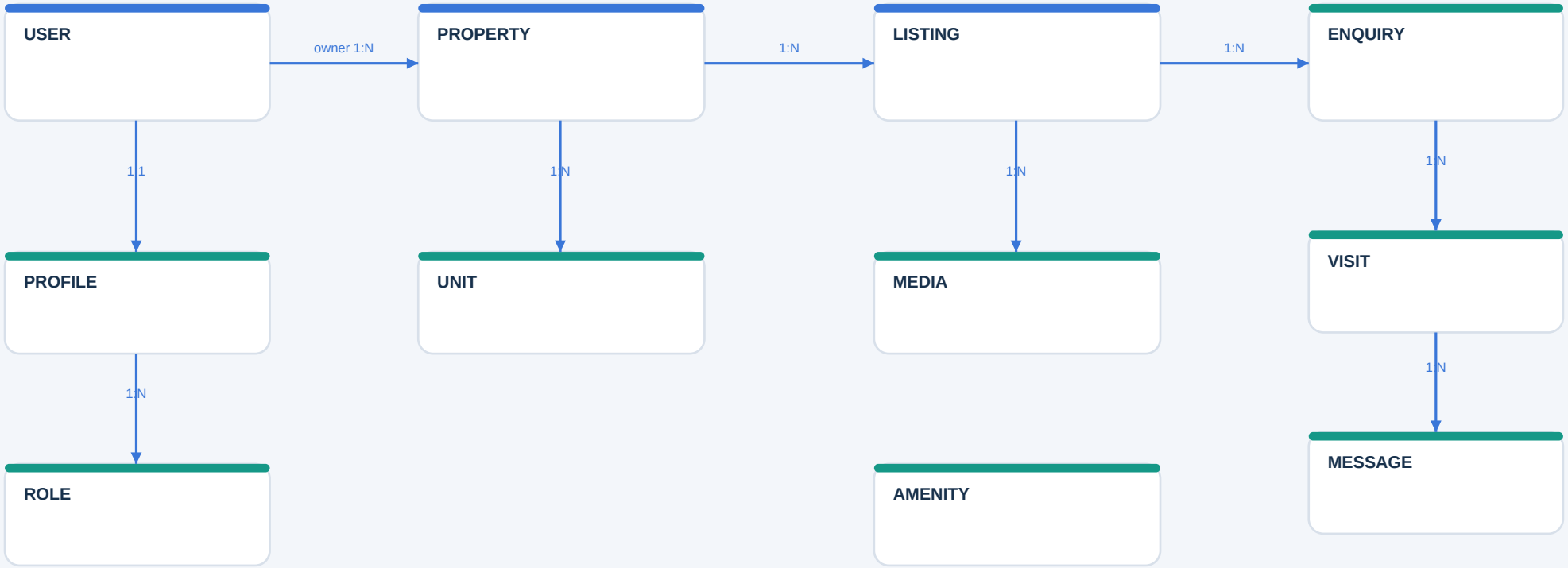
A visit is not confirmed until the lister accepts a slot.



Use idempotency keys for request creation. Record actor and reason for cancel/reschedule/no-show. Keep block/report available in contact surfaces.

11 — DATA DESIGN: conceptual ERD

Physical property, rentable inventory and public offer are separate entities.



Supporting tables: favourite, saved_search, verification_case, moderation_case, report, review, notification, audit_event, outbox_event. Amenity is N:M through listing_amenity.

12 — DATA DESIGN: supporting data domains & ownership

Entity families are grouped by domain; foreign keys enforce integrity and module ownership controls change.

IDENTITY

app_user • user_role • user_profile | Account, roles, consent and profile

CATALOG

property • unit • listing • listing_media | Physical place, rentable inventory and offer

DISCOVERY

amenity • listing_amenity • favourite • saved_search | Search facets and seeker intent

ENGAGEMENT

enquiry • conversation • message • visit_request | Lead, communication and visit lifecycle

TRUST

verification_case • authority_evidence • moderation_case • report • review | Evidence, decisions and complaints

PLATFORM

notification • audit_event • outbox_event • idempotency_record | Delivery, traceability and async work

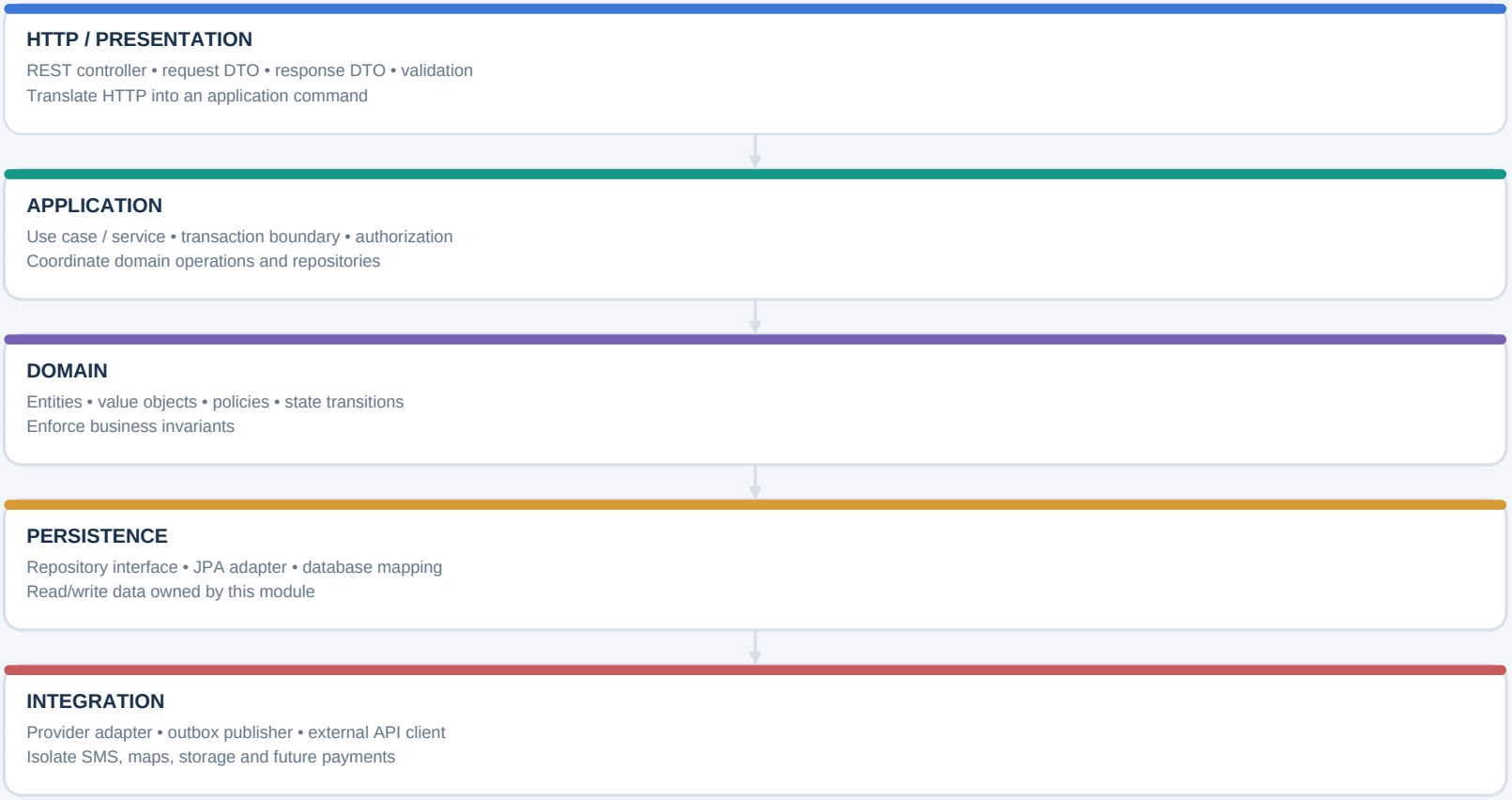
FUTURE COMMERCE

booking • payment_attempt • payment_event • refund • ledger_entry | Booking intent and payment settlement are separate

Data rules: UUID IDs, timestamptz instants, integer paise for money, FK/unique/check constraints, Flyway migrations, explicit retention and purge policy.

13 — LLD: inside a Spring Boot module

HTTP controllers translate requests; business rules live in application and domain layers.



Example: SubmitListing → check permission → validate completeness → persist SUBMITTED → write ListingSubmitted outbox event → return status.

14 — LLD: API surface map

Organise endpoints by resource and capability; final payload contracts are maintained in OpenAPI.

AUTH

POST /auth/otp • POST /auth/verify • POST /auth/refresh

Login and session lifecycle

PROFILE

GET/PATCH /me • PATCH /me/preferences

Profile and consent

LISTINGS

POST /listings • PATCH /listings/{id} • POST /listings/{id}/submit

Draft and publish workflow

DISCOVERY

GET /listings • GET /listings/{id} • GET /localities

Search and listing detail

ENGAGEMENT

POST /listings/{id}/enquiries • POST /visits • PATCH /visits/{id}

Enquiry and visit scheduling

TRUST / ADMIN

POST /reports • GET /admin/review-queue • POST /admin/listings/{id}/decision

Safety and moderation

MEDIA

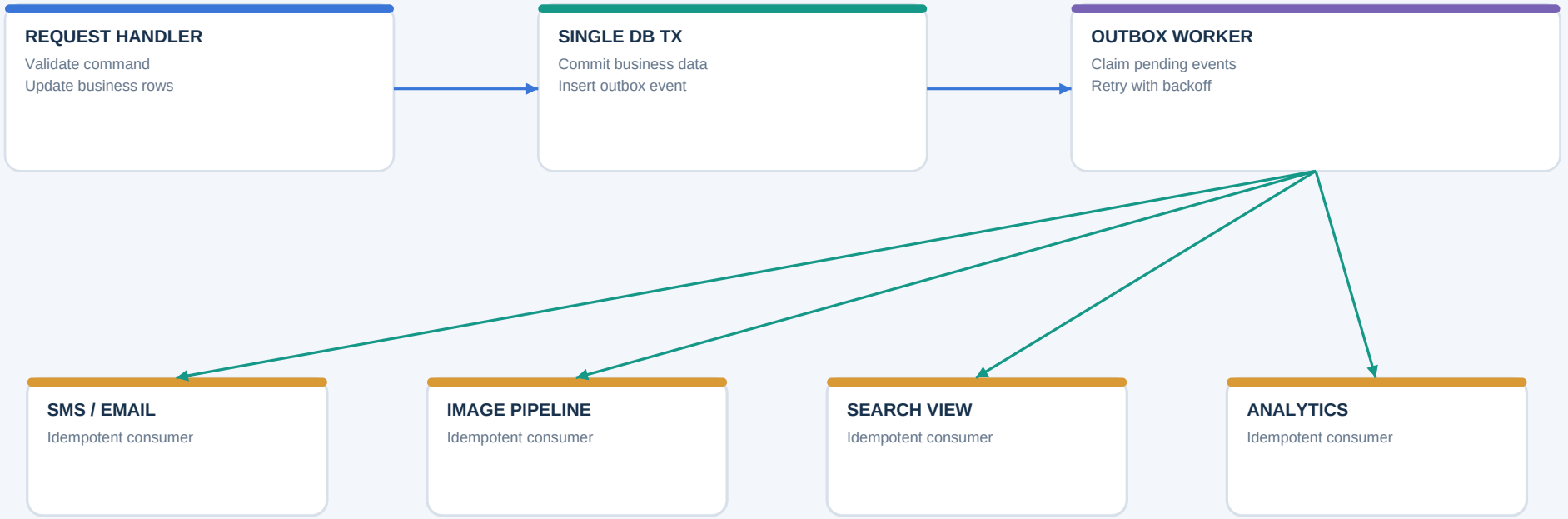
POST /media/upload-intents • POST /media/{id}/complete

Signed upload workflow

API rules: version contracts, validate server-side, enforce object-level authorization, paginate, return stable errors and use idempotency for retryable writes.

15 — LLD: asynchronous processing & transactional outbox

Business state and the event record commit together; a worker delivers side effects with retries.

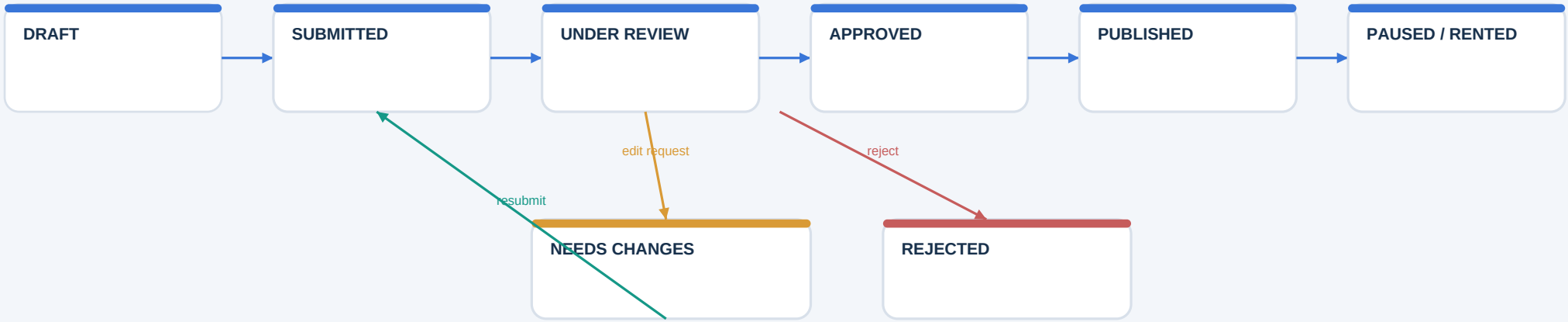


Do not send notifications before commit. Consumers tolerate duplicate delivery; failures move to a monitored retry / dead-letter state.

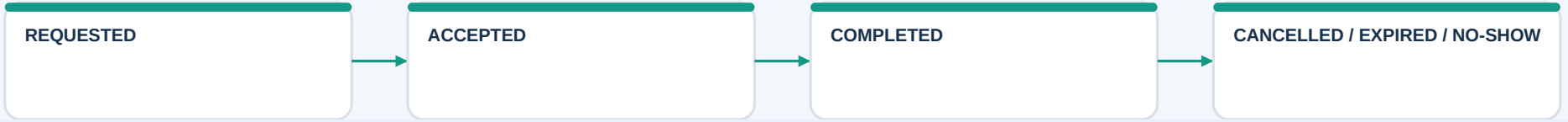
16 — LLD: state machines for listing and visit

Explicit states prevent accidental publication and ambiguous support decisions.

LISTING LIFECYCLE



VISIT REQUEST LIFECYCLE



Search visibility: only PUBLISHED + available + not expired + not suspended. Every transition is validated and records actor/time.

17 — TRUST & SAFETY: separate evidence claims

Never collapse phone, identity, authority, listing review and visit completion into one generic badge.

PHONE VERIFIED Controls a phone number	**LIMITATION** Does not prove identity or authority
IDENTITY CHECKED Identity claim checked via eligible route	**LIMITATION** Does not prove ownership or future behaviour
AUTHORITY CHECKED Evidence supports right to list	**LIMITATION** Does not guarantee title or room condition
LISTING REVIEWED Content checked against platform policy	**LIMITATION** Does not mean physical inspection
VISIT COMPLETED Participants confirm a visit	**LIMITATION** Does not prove lease or payment

Identity-document integrations require legal eligibility, approved provider route, explicit consent, minimised data and a retention/deletion policy.

18 — PRODUCTION: security & privacy boundaries

Classify data, limit access, minimise collection and audit sensitive actions.

PUBLIC
Published listing fields
Approximate location • price • amenities

AUTHENTICATED
Profile • enquiry • visits
Only authorised user and resource owner

RESTRICTED
Exact address • evidence
Purpose-bound access • masked by default

SECRET
OTP hashes • tokens • API keys
Secret manager • never log • rotate

AUTHN
Short-lived sessions • OTP throttling

AUTHZ
RBAC + ownership checks

DATA
TLS • encryption • retention

APP
Validation • encoding • upload scan

OPS
Audit • secret rotation • backups

Exact house number / door pin is not public by default. Mask identity evidence. Never log OTPs, tokens, identity documents or payment credentials.

19 — PRODUCTION: non-functional requirements & observability

Define measurable objectives before launch; measure the complete user journey, not only uptime.

PERFORMANCE

Initial target: search p95 < 700 ms; listing detail p95 < 1 s, excluding external map latency.

AVAILABILITY

Initial target: 99.5% monthly for core browse/search; degrade gracefully if maps or messaging fail.

SCALABILITY

Stateless API replicas; DB indexes and connection pool limits; cache only measured hot reads.

RELIABILITY

Timeouts, bounded retries, idempotency, outbox, backup and restore drills.

ACCESSIBILITY

WCAG 2.2 AA target; keyboard, screen-reader labels, contrast and responsive design.

OBSERVABILITY

Request IDs, structured logs, RED metrics, traces, funnel analytics and alert thresholds.

Targets are initial engineering objectives, not measured guarantees. Load-test with realistic search patterns and revise from pilot telemetry.

20 — PRODUCTION: quality strategy & release gates

Test business invariants at multiple levels; highest-risk user journeys get explicit release gates.

UNIT

Domain rules • pricing • state transitions • permission policies

INTEGRATION

Repositories • PostGIS • transactions • outbox • provider adapters

API / CONTRACT

OpenAPI • auth scopes • error schema • pagination • idempotency

END-TO-END

OTP → search → listing → enquiry → visit → report / moderation

SECURITY

Access control • upload abuse • rate limits • secret scanning

OPERATIONS

Migration rehearsal • backup restore • alert test • rollback drill

Pilot gate: no critical security defects; no unpublished listing leakage; ownership checks enforced; visit states consistent; restore tested; moderation staffed.

21 — DELIVERY: build sequence from foundation to pilot

Implement vertical slices in dependency order; every phase ends with a working testable flow.

FOUNDATION
Repo • CI • environments • DB migrations

IDENTITY
OTP • sessions • profile • roles • audit

CATALOG
Property • unit • listing draft • media

TRUST
Authority evidence • moderation • publish

DISCOVERY
Search • filters • geo • detail • favourites

ENGAGEMENT
Enquiry • visits • notifications • reports

HARDEN
Security • accessibility • performance • restore

PILOT
Seed inventory • invite users • support • metrics

Pilot metrics: real available inventory, listing freshness, search-to-enquiry conversion, enquiry-to-visit conversion, response time and report rate.

22 — ARCHITECTURE DECISIONS & rationale

Keep the first release understandable, secure and affordable to operate.

MODULAR MONOLITH

Fast pilot; clear boundaries without distributed-system overhead.

POSTGRES + POSTGIS

Relational integrity and location search in one source of truth.

PROPERTY ≠ UNIT ≠ LISTING

Place, inventory and public price/availability change independently.

OUTBOX FOR SIDE EFFECTS

Avoid lost notifications when DB commit succeeds but provider call fails.

HUMAN MODERATION

Trust decisions need evidence, reason codes, review and appeal.

PAYMENTS LATER

Start with discovery and visits; add hosted checkout after provider/legal review.

Extract a service only when independent scaling, team ownership, release cadence or fault isolation is demonstrated by metrics.

23 — GLOSSARY: shared engineering vocabulary

A common language for product, design, frontend, backend, QA and operations.

HLD
High-Level Design: system context, major components, boundaries and deployment.

LLD
Low-Level Design: module internals, APIs, data contracts, sequences and state rules.

ERD
Entity Relationship Diagram: data entities and relationships/cardinalities.

RBAC
Role-Based Access Control: permissions assigned to roles and checked per resource.

POSTGIS
PostgreSQL spatial extension for coordinates, radius and nearby queries.

OUTBOX
Table that records events in the same transaction as business data.

IDEMPOTENCY
Repeated request/event produces one logical outcome rather than duplicates.

P95
95th percentile latency: 95% of requests complete at or below this time.

WAF
Web Application Firewall: filters malicious web traffic at the edge.

ADR
Architecture Decision Record: decision, context, options and trade-offs.

This is the recommended baseline, not a vendor-specific final deployment. Pair it with the detailed product requirements and the visual product-flow pack.